

PEKAN BARU SIMPANGTIGA, Indonesia

WMO: 961090

Lat: 0.461N

Lon: 101.445E

Elev: 31

StdP: 100.95

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.1	(c) 22.5	(d) 20.1	(e) 14.8	(f) 29.3	(g) 20.9	(h) 15.6	(i) 28.2	(j) 6.4	(k) 28.4	(l) 5.5	(m) 29.4	(n) 0.6	(o) 210	(p) 0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.5	34.6	26.2	34.1	26.2	33.6	26.2	27.9	32.6	27.5	32.1	27.2	31.7	1.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.2	30.9	26.2	21.7	30.4	25.9	21.3	30.1	89.4	32.9	87.6	32.3	86.0	31.9	34.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.5	(b) 4.8	(c) 4.2	(e) 21.0	(f) 35.8	(g) 0.8	(h) 0.8	(i) 20.4	(j) 36.4	(k) 19.9	(l) 36.9	(m) 19.4	(n) 37.3	(o) 18.9	(p) 37.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			19.7	29.7	1.4	1.6	18.7	30.8	17.9	31.8	17.1	32.7	16.1	33.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.3	27.7	28.1	28.5	28.7	28.6	28.2	28.1	27.9	28.0	27.8	27.5
	DBStd	1.09	1.07	0.99	1.04	1.01	1.08	1.02	1.02	1.04	1.00	0.97	0.84	0.99
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6579	537	495	562	556	578	557	565	562	536	558	533	541
	CDD18.3	3537	278	262	303	306	320	307	307	303	286	300	283	283
	CDH23.3	36030	2554	2548	3069	3218	3542	3438	3310	3196	2865	2953	2709	2627
	CDH26.7	14552	878	946	1254	1371	1532	1516	1418	1346	1152	1189	1029	920
(14) Wind	WSAvg	1.0	1.1	0.9	0.9	0.9	0.9	1.1	1.2	1.3	1.1	0.9	1.0	1.1
(15) Precipitation	PrecAvg	2603	189	198	252	261	225	162	126	175	217	239	317	268
	PrecMax	3262	499	531	460	419	383	368	260	417	381	443	616	554
	PrecMin	1433	42	31	79	105	45	30	33	34	14	65	113	105
	PrecStd	376	101	122	104	84	91	79	63	99	94	97	108	94
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.9	34.6	35.1	35.1	35.0	34.4	34.4	34.2	34.6	33.9	33.4
		MCWB	25.1	25.9	26.1	26.7	26.8	26.2	26.0	25.8	26.1	25.6	26.1	25.5
	2%	DB	32.7	33.0	33.8	34.1	34.1	34.1	33.8	33.8	33.4	33.5	33.0	32.7
		MCWB	25.4	25.7	26.0	26.6	26.8	26.3	26.0	25.9	26.1	25.9	25.9	25.6
	5%	DB	31.8	32.1	33.0	33.2	33.4	33.3	33.1	33.0	32.7	32.8	32.2	31.9
		MCWB	25.4	25.6	26.0	26.5	26.7	26.3	25.9	25.9	26.0	26.0	25.9	25.6
	10%	DB	30.9	31.2	32.1	32.5	32.7	32.7	32.2	32.1	31.9	31.9	31.5	31.0
		MCWB	25.4	25.5	26.0	26.4	26.7	26.3	25.9	25.8	25.9	25.9	25.9	25.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.2	27.9	28.1	28.4	28.1	27.6	27.5	27.6	27.7	27.6	27.0
		MCDB	31.2	31.7	32.4	33.0	33.2	32.9	32.3	32.3	32.4	32.4	31.8	31.0
	2%	WB	26.4	26.6	27.1	27.5	27.8	27.5	27.0	26.8	27.0	27.0	26.8	26.5
		MCDB	30.4	30.9	31.7	32.3	32.4	32.3	31.8	31.7	31.8	31.6	31.0	30.4
	5%	WB	26.1	26.2	26.7	27.1	27.3	27.1	26.6	26.5	26.6	26.6	26.5	26.2
		MCDB	29.9	30.4	31.1	31.7	31.7	31.7	31.3	31.3	31.2	31.0	30.7	30.0
	10%	WB	25.7	25.8	26.3	26.7	26.9	26.6	26.2	26.2	26.2	26.2	26.1	25.9
		MCDB	29.3	29.9	30.6	31.1	31.2	31.1	30.8	30.8	30.6	30.5	30.3	29.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.0	8.6	8.7	8.5	8.7	8.8	8.7	8.4	8.3	8.1	7.5
		MCDBR	8.9	9.1	9.7	9.6	9.5	9.6	9.6	9.7	9.6	9.4	9.0	8.8
		MCWBR	3.1	3.1	3.5	3.4	3.4	3.5	3.3	3.4	3.6	3.4	3.3	3.1
	5% WB	MCDBR	7.6	8.2	8.9	8.9	8.8	8.9	8.9	8.9	8.9	8.6	8.2	7.8
		MCWBR	3.1	3.3	3.7	3.6	3.7	3.7	3.6	3.7	3.9	3.6	3.4	3.1
(40) Clear-Sky Solar Irradiance	taub		0.477	0.592	0.556	0.473	0.468	0.495	0.488	0.554	0.593	0.619	0.476	0.443
	taud		2.229	1.887	1.999	2.250	2.271	2.182	2.206	2.003	1.900	1.820	2.244	2.351
	Ebn at Noon		856	766	790	839	820	784	796	762	750	736	850	882
	Edh at Noon		147	210	187	141	134	143	141	178	203	221	144	128
(44) All-Sky Solar Radiation	RadAvg		4.35	4.52	4.89	5.01	4.91	4.81	4.81	4.77	4.68	4.73	4.68	4.36
	RadStd		0.29	0.23	0.31	0.13	0.20	0.19	0.19	0.27	0.43	0.60	0.17	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.32	+0.34	N/A	+0.32	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page